

QUADRATIC EQUATION

An equation of second degree is called a quadratic equation.

It is of the form

$$ax^2 + bx + c = 0$$

The roots of quadratic equation are

$$x = \frac{-b \pm (b^2 - 4ac)^{1/2}}{2a}$$

BINOMIAL THEOREM

If n is any integer, or a fraction and x is any real number, then

$$(1 + x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \dots$$

If $|x| \ll 1$, then $(1 + x)^n \approx 1 + nx$.

MENSURATION

★ Area of circle = $\pi r^2 =$

$$\frac{\pi D^2}{4} \quad \left(\because r = \frac{D}{2} \right)$$

★ Surface area of a sphere = $4\pi r^2 = \pi D^2$

★ Volume of sphere = $\frac{4}{3}\pi r^3$

★ Surface area of a cylinder = $2\pi r(r + h)$

★ Volume of a cylinder = $\pi r^2 h$

★ Curved surface area of a cone = πrl

★ Volume of cone = $\frac{1}{3}\pi r^2 h$

★ Surface area of a cube = $6(\text{side})^2$

★ Volume of a cube = $(\text{side})^3$

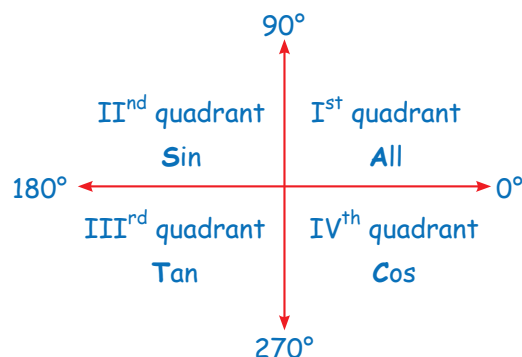
FOUR QUADRANTS

★ Ist quadrants $\Rightarrow$ All trigonometric ratios are positive

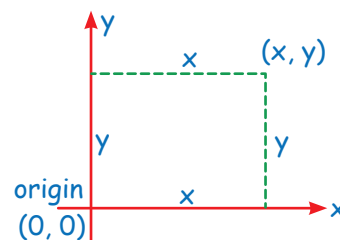
★ IInd quadrants $\Rightarrow$ Only $\sin \theta$ and $\operatorname{cosec} \theta$ are positive

★ IIIrd quadrants $\Rightarrow$ Only $\tan \theta$ and $\cot \theta$ are positive

★ IVth quadrants $\Rightarrow$ Only $\cos \theta$ and $\sec \theta$ are positive



POSITION OF A POINT IN xy PLANE



DISTANCE FORMULA

★ The distance (d) between two points $P(x_1, y_1)$ and

$$Q(x_2, y_2) \text{ is given by } d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

FUNDAMENTAL TRIGONOMETRIC RELATIONS

- ★ $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$
- ★ $\sec \theta = \frac{1}{\cos \theta}$
- ★ $\cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta}$
- ★ $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- ★ $\sin^2 \theta + \cos^2 \theta = 1$
- ★ $1 + \tan^2 \theta = \sec^2 \theta$
- ★ $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
- ★ $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- ★ $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- ★ $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- ★ $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- ★ $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- ★ $\sin 2A = 2 \sin A \cos A$
- ★ $\cos 2A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A = \cos^2 A - \sin^2 A$
- ★ $\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$
- ★ $\sin A - \sin B = 2 \sin \left(\frac{A-B}{2} \right) \cos \left(\frac{A+B}{2} \right)$
- ★ $\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$
- ★ $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$

LOGARITHMIC FORMULAS

- ★ $\log_a mn = \log_a m + \log_a n$
- ★ $\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$
- ★ $\log_a m = \log_b m \times \log_a b$
- ★ $\log_a b^c = c \log_a b$
- ★ $\log_a 1 = 0$
- ★ $\log_a a = 1$
- ★ $x = e^{\log x}$

APPROXIMATE VALUES

If angle (θ) is small then

- ★ $\sin \theta \approx \theta$
- ★ $\cos \theta \approx 1$
- ★ $\tan \theta \approx \theta$
- ★ $\sin \theta \approx \tan \theta$

DIFFERENTIAL FORMULAS

- ★ $\frac{dc}{dx} = 0$, [c = constant]
- ★ $\frac{d(cy)}{dx} = c \frac{dy}{dx}$
- ★ $\frac{d(x^n)}{dx} = nx^{n-1}$
- ★ $\frac{d[f(x) \pm g(x)]}{dx} = \frac{df(x)}{dx} \pm \frac{dg(x)}{dx}$
- ★ $\frac{d\{f(x)g(x)\}}{dx} = f(x) \frac{dg(x)}{dx} + g(x) \frac{df(x)}{dx}$
- ★ $\frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx}}{\{g(x)\}^2}$
- ★ $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$
- ★ $\frac{d}{dx} e^x = e^x$
- ★ $\frac{d}{dx} u^n = nu^{n-1} \frac{du}{dx}$
- ★ $\frac{d}{dx} \log x = \frac{1}{x}$
- ★ $\frac{d}{dx} (\sin x) = \cos x$
- ★ $\frac{d}{dx} (\cos x) = -\sin x$

$$\star \frac{d}{dx}(\tan x) = \sec^2 x$$

$$\star \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$\star \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\star \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$\star \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

$$\star \frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

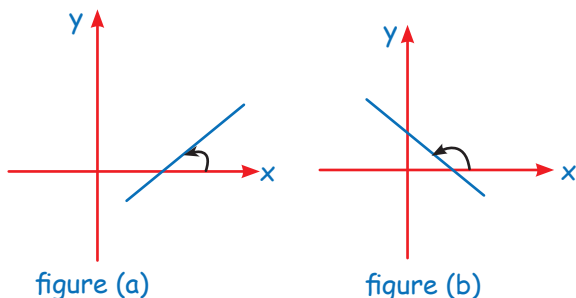
SLOPE OF A LINE

It is the tan of angle made by a line with the positive direction of x-axis, measured in anticlockwise direction.

★ Slope = $\tan \theta$

★ In figure (a) slope is positive

★ In figure (b) slope is negative



MAXIMA AND MINIMA

For a function $y = f(x)$, find critical points first by:

$$\frac{dy}{dx} = 0$$

Then at those points:

★ Condition for Minima: $\frac{d^2y}{dx^2} > 0$

★ Condition for Maxima: $\frac{d^2y}{dx^2} < 0$

INTEGRAL FORMULAS

$$\star \int dx = x + c \quad \text{where } c = \text{constant}$$

$$\star \int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\star \int \left(\frac{1}{x}\right) dx = \log_e x + c$$

$$\star \int \sin x dx = -\cos x + c$$

$$\star \int \sin ax dx = -\frac{\cos ax}{a} + c$$

$$\star \int \cos x dx = \sin x + c$$

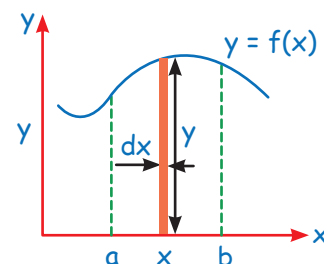
$$\star \int \sec^2 x dx = \tan x + c$$

$$\star \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$\star \int \tan x dx = \log |\sec x| + c$$

$$\star \int e^x dx = e^x + c$$

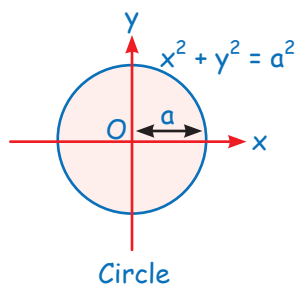
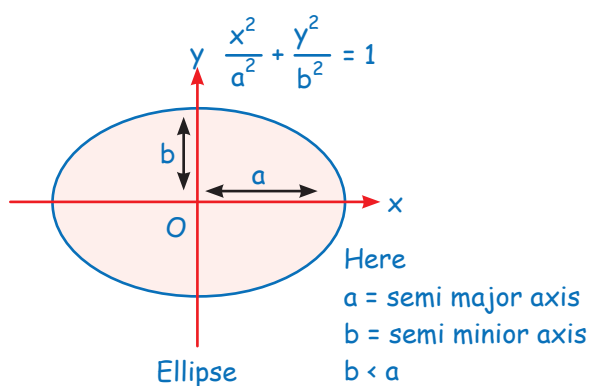
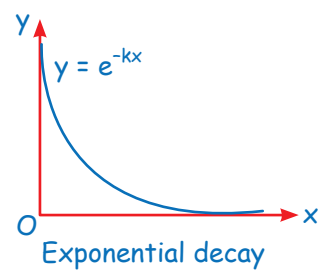
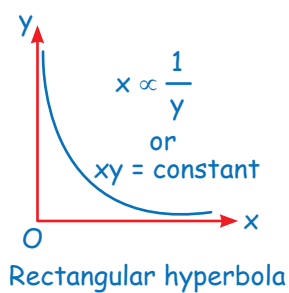
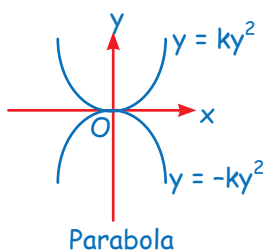
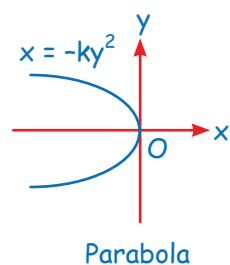
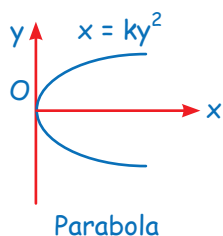
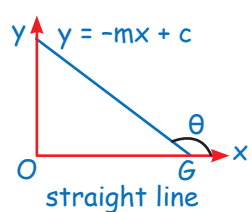
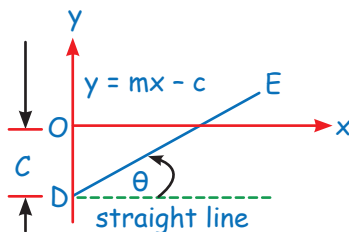
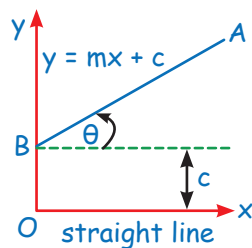
AREA UNDER A CURVE AND DEFINITE INTEGRATION



$$\star \int_a^b f(x) dx = \text{Area under the curve and x-axis}$$

[Where $\int_a^b f(x) dx$: Definite integral, a: Lower limit, b: Upper limit]

SOME STANDARD GRAPHS AND THEIR EQUATIONS



TRIGONOMETRY

Q. $\sin(90^\circ - \theta)$ is equal to:

Sol. $\sin(90^\circ - \theta) = \cos \theta$

Q. 1 radian is equal to:

Sol. π radian = 180°

$$1 \text{ radian} = \frac{180}{\pi} \text{ degrees}$$

Q. $\sin 100\pi$ is equal to:

Sol. $\sin n\pi = 0, n = 0, 1, 2, \dots$

$$\therefore \sin 100\pi = 0$$

Q. $\cos(180^\circ - \theta)$ is equal to:

Sol. $\cos(180^\circ - \theta) = -\cos \theta$

Q. $\cos 2A$ is equal to:

Sol. $\cos 2A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A = \cos^2 A - \sin^2 A$

Q. $\sin^2 4\theta + \cos^2 4\theta$ is equal to:

Sol. $\sin^2(x) + \cos^2(x) = 1$

$$\therefore \sin^2(4\theta) + \cos^2(4\theta) = 1$$

Q. $\sin 300^\circ$ is equal to:

Sol. $\sin(270^\circ + \theta) = -\cos \theta$

$$\begin{aligned} \sin(300^\circ) &= \sin(270^\circ + 30^\circ) = -\cos 30^\circ \\ &= -\frac{\sqrt{3}}{2} \end{aligned}$$

Q. Evaluate:

$$\begin{aligned} \text{Sol. (1) } \cos 15^\circ &= \cos(45^\circ - 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}} \end{aligned}$$

$$(2) \cos 53^\circ = \frac{3}{5}$$

$$(3) \tan 37^\circ = \frac{\sin 37^\circ}{\cos 37^\circ} = \frac{\frac{3}{5}}{\frac{4}{5}} = \frac{3}{4}$$

$$(4) \sin 53^\circ - \cos 37^\circ = \frac{4}{5} - \frac{4}{5} = 0$$

Q. Evaluate:

$$\text{Sol. (1) } 2 \sin 15^\circ \cos 15^\circ = \sin 30^\circ = \frac{1}{2}$$

$$(2) \sin 22.5^\circ \cos 22.5^\circ = \frac{1}{2} \sin 45^\circ = \frac{1}{2\sqrt{2}}$$

$$(3) \tan 75^\circ = \tan(45^\circ + 30^\circ)$$

$$= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$(4) \sin^2 22.5^\circ = \frac{1 - \cos 45^\circ}{2} = \frac{1 - \frac{1}{\sqrt{2}}}{2} = \frac{\sqrt{2}-1}{2\sqrt{2}}$$

Q. Evaluate:

$$\text{Sol. (1) } \cos\left(\frac{5\pi}{4}\right) = \cos\left(\pi + \frac{\pi}{4}\right) = -\cos\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$(2) \sin\left(\frac{2\pi}{3}\right) = \sin\left(\pi - \frac{\pi}{3}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$(3) \sin\left(\frac{3\pi}{4}\right) = \sin\left(\pi - \frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$(4) \tan\left(\frac{7\pi}{6}\right) = \tan\left(\pi + \frac{\pi}{6}\right) = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

Q. $\sqrt{1 + \sin \theta}$ is equal to

$$\text{Sol. } \sqrt{1 + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$(\because \sin 2\theta = 2 \sin \theta \cos \theta)$$

$$= \sqrt{\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$= \sqrt{\left(\sin \frac{\theta}{2} + \cos \frac{\theta}{2}\right)^2} = \sin \frac{\theta}{2} + \cos \frac{\theta}{2}$$

Q. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = \frac{7}{3}$, then $\tan \theta =$

Sol. $3 \sin \theta + 3 \cos \theta = 7 \sin \theta - 7 \cos \theta$

$$\Rightarrow -4 \sin \theta = -10 \cos \theta \Rightarrow \tan \theta = \frac{5}{2}$$

ALGEBRA

Q. The roots of $100x^2 - 20x + 1 = 0$ is:

Sol. $100x^2 - 20x + 1 = 0$

$$\Rightarrow (10x-1)^2 = 0$$

$$\therefore \text{two roots are } \frac{1}{10}, \frac{1}{10}$$

Q. If $\frac{1}{2}$ is a root of the quadratic equation $x^2 - mx - \frac{5}{4} = 0$, then value of m is:

$$\text{Sol. } \left(\frac{1}{2}\right)^2 - m\left(\frac{1}{2}\right) - \frac{5}{4} = 0 \Rightarrow m = -2$$

Q. Find the sum of first 20 natural Number.

Sol. Sum of n natural numbers $= \frac{n(n+1)}{2}$
 $= \frac{20}{2}(20+1) = 10 \times 21 = 210$

Q. The sum of the first five multiples of 3 is:

Sol. The first five multiples of 3 is 3, 6, 9, 12 and 15
 $a = 3$ and $d = 3$
 $n = 5$

$$\text{Sum, } S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_5 = \frac{5}{2}[2(3) + (5-1)3] = \frac{5}{2}[6 + 12] = \frac{5}{2}[18]$$

$$= 5 \times 9 = 45$$

Q. Find sum of $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} \dots \infty$.

Sol. This series is G.P. of common ratio $r = -\frac{1}{2}$

$$S_\infty = \frac{a}{1-r} = \frac{1}{1 - \left(-\frac{1}{2}\right)} = \frac{1}{1 + \frac{1}{2}} = \frac{2}{3}$$

Q. Find sum of $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$ up to ∞ .

Sol. This series is G.P. of common ratio $= \frac{1}{3}$

$$S_\infty = \frac{a}{1-r} = \frac{1}{1 - \frac{1}{3}} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$$

Q. The sum of $1 + 3 + 5 + 7 + \dots$ upto n terms is:

Sol. $a = 1, d = 2$, number of terms $= n$

$$\therefore S = \frac{n}{2}[2(1) + (n-1)2] = n^2$$

Q. Find approximate value of the $\sqrt{0.95}$:

Sol. $(0.95)^{1/2} = (1 - 0.05)^{1/2}$

Binomial expression, if $x \ll 1$, then $(1 + x)^n = (1 + nx)$

$$= \left(1 - \frac{0.05}{2}\right) = \left(1 - \frac{5}{200}\right) = \left(1 - \frac{1}{40}\right)$$

$$= (1 - 0.025) = 0.975$$

Q. Find approximate value of the $\sqrt{104}$:

Sol. $(100 + 4)^{1/2}$

$$= \left[100 \left(1 + \frac{4}{100}\right)\right]^{1/2} = 10(1.02) = 10.2$$

Q. Find approximate value of the $(4.04)^3$

Sol. $(4 + 0.04)^3$

$$\left(4 + \frac{4}{100}\right)^3 = \left[4 \left(1 + \frac{1}{100}\right)\right]^3$$

$$= 4^3 \times \left(1 + \frac{3}{100}\right) = 64 \times \frac{103}{100} = 65.92$$

Q. Find approximate value of the $(9.6)^4$

Sol. $(10 - 0.4)^4$

$$= \left(10 - \frac{4}{10}\right)^4 = \left[10 \left(1 - \frac{4}{100}\right)\right]^4 = 10^4 \left(1 - \frac{16}{100}\right)$$

$$= 10^4 (1 - 0.16) = 10^4 \times 0.84 = 8400$$

Q. $\log 25 + \log 4 - \log 5$ is equal to

Sol. $\log 5^2 + \log 2^2 - \log 5$
 $= 2 \log 5 + \log 2^2 - \log 5 \quad (\because \log m^n = n \log m)$
 $\log (5 \times 4) = \log 20$
 $(\because \log (m \times n) = \log m + \log n)$

Q. $\log_e 8$ is equal to

Sol. $\log 2^3 = 3 \log 2$

Q. $\log_e 15$ is equal to

Sol. $\log (3 \times 5) = \log 3 + \log 5$

Q. $\log_5 x - \log_5(y) = 2$, find the value of $\frac{x}{y}$

Sol. $\log_5 \left(\frac{x}{y}\right) = \frac{2}{1} \Rightarrow \frac{x}{y} = 5^2 = 25$

CALCULUS

Q. If $y = \sin^2 x - 2 \tan^2 x$, then $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$ is:

Sol. $\frac{dy}{dx} = 2 \sin x \cos x - 4 \tan x \cdot \sec^2 x$
 $= 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} - 4 \times 1 \times \sqrt{2} \times \sqrt{2} = 1 - 8 = -7$

Q. If $y = x^3 + 2x + 1$ then $\frac{dy}{dx}$ at $x = 1$ is:

Sol. $\frac{dy}{dx} = 3x^2 + 2$
 $f(x = 1) = 3 \times 1^2 + 2 = 5$

Q. $y = \frac{1+x}{e^x}$ then $\frac{dy}{dx}$ is equal to:

Sol. $\frac{dy}{dx} = \frac{e^x - (1+x)e^x}{e^x \cdot e^x} = \frac{-x}{e^x}$

Q. $\frac{d}{dr}(\pi r^2)$ is equal to:

Sol. $\frac{d}{dr}(\pi r^2) = 2\pi r$

Q. If $y = x^2 + x - 1$ then $\frac{dy}{dx}$ at $x = 1$ is equal to:

Sol. $y = x^2 + x + 1$

$$\frac{dy}{dx} = 2x + 1$$

$$\text{At } x = 1, \frac{dy}{dx} = 2 \times 1 + 1 = 3$$

Q. Find derivative of $y = x \sin x$ with respect to independent variable x .

$$y = x \sin x$$

Sol. $y = x \sin x$

$$dy/dx = x \cos x + \sin x$$

Q. Find derivative of $y = e^x \ln x$ with respect to x .

Sol. $e^x \ln x + \frac{e^x}{x}$

Q. Find derivative of $y = (x-1)(x^2 + x + 1)$ with respect to independent variable x .

Sol. $\frac{dy}{dx} = (x-1)(2x+1) + (x^2 + x + 1) \cdot 1 = 3x^2$

Q. If $y = a(1 - \cos \theta)$ and $x = a(\theta + \sin \theta)$, then find the slope of tangent of y verses x at $\theta = \frac{\pi}{2}$.

Sol. $\frac{dy}{d\theta} = a \sin \theta, \frac{dx}{d\theta} = a(1 + \cos \theta)$

$$\text{Slope, } \frac{dy}{dx} = \frac{a \sin \theta}{a(1 + \cos \theta)} = \frac{\sin \frac{\pi}{2}}{1 + \cos \frac{\pi}{2}} = 1$$

Q. The slope of the normal to the curve $y = x^2 - \frac{1}{x^2}$ at $(-1, 0)$ is:

Sol. Slope, of Tangent, $\frac{dy}{dx} = 2x - \frac{(-2)}{x^3} = 2x + \frac{2}{x^3}$

at point $(-1, 0)$

$$\frac{dy}{dx} = 2 \times (-1) + \frac{2}{(-1)^3} = -2 - \frac{2}{1} = -4$$

$$\text{Slope of normal} = \frac{-1}{\text{Slope of Tangent}} = \frac{-1}{-4} = \frac{1}{4}$$

Q. $y = \sin u, u = \cos x$. Find $\frac{dy}{dx}$.

Sol. $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = -\cos u \cdot \sin x$

Q. $y = 3t^2 - 1, x = t^2$. Find $\frac{dy}{dx}$.

Sol. $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{6t}{2t} = 3$

Q. The value of $\int_0^{\pi/2} \sin 2\theta d\theta$ is:

Sol. $\int_0^{\pi/2} \sin 2\theta d\theta = \left[\frac{-\cos 2\theta}{2} \right]_0^{\pi/2} = -\left[\frac{\cos \pi - \cos 0}{2} \right]$
 $= -\left[\frac{-1 - 1}{2} \right] = 1$

Q. Value of $\int_0^1 \frac{1}{(3-2x)^2} dx$ is:

Sol. $\int_0^1 (3-2x)^{-2} dx = \left[\frac{(3-2x)^{-2+1}}{(-2+1) \times -2} \right]_0^1$
 $= \frac{1}{2} \left(1 - \frac{1}{3} \right) = \frac{1}{3}$

Q. $\int (4 \cos t + t^2) dt$ is equal to:

Sol. $\int (4 \cos t + t^2) dt = \int 4 \cos t dt + \int t^2 dt$
 $= 4 \sin t + \frac{t^3}{3} + C$

Q. $\int \frac{1}{\sqrt{x}} dx$ is

Sol. $\int x^{\left(-\frac{1}{2}\right)} dx = \frac{x^{\left(-\frac{1}{2}+1\right)}}{\frac{1}{2}}$
 $= 2\sqrt{x} + C$

Q. If $y = \sin(2x^2)$, then $\frac{dy}{dx}$ is:

Sol. $\frac{dy}{dx} = \frac{d\sin(2x^2)}{d(2x^2)} \times \frac{d(2x^2)}{dx} = \cos(2x^2) \times 4x$

Q. Maximum and minimum values of function $2x^3 - 15x^2 + 36x + 11$ is

Sol. For maxima and minima, $dy/dx = 0$

$$6x^2 - 30x + 36 = 0$$

$$x^2 - 5x + 6 = 0 \Rightarrow x = 2, 3$$

$$\text{At } x = 2, y = 39 \quad (\text{max.})$$

$$\text{At } x = 3, y = 38 \quad (\text{min.})$$

Q. $\int \left(\frac{3}{2\sqrt{x}} \right) dx$ will be

Sol. $3\sqrt{x} + C$

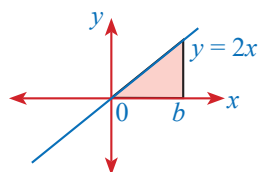
Q. $\int \left(\frac{1}{3\sqrt[3]{x}} \right) dx$ will be?

Sol. $\frac{x^{\frac{2}{3}}}{\frac{2}{3}} + C$

Q. $\int_{-4}^{-1} \frac{\pi}{2} d\theta$ will be?

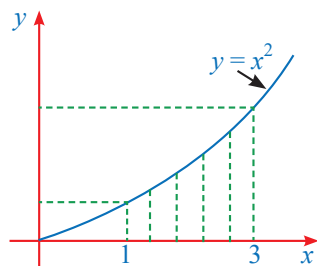
Sol. $\frac{\pi}{2} [\theta]_{-4}^{-1} = \frac{\pi}{2} (-1 + 4) = \frac{3\pi}{2}$

Q. Use a definite integral to find the area of the region between the given curve $y = 2x$ and the x-axis on the interval $[0, b]$



Sol. Area = $\int_0^b 2x dx = b^2$

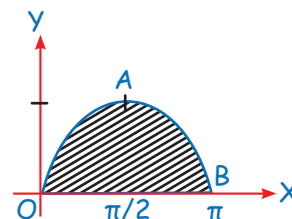
Q. For function $y = f(x) = x^2$, find average value of y from $x = 1$ to $x = 3$



Sol. $y_{\text{avg}} = \frac{\int_1^3 x^2 dx}{3-1} = \frac{1}{2} \left[\frac{x^3}{3} \right]_1^3 = \frac{13}{3}$

Q. Find the area of the curve $y = \sin x$ between 0 and π .

Sol. Given $y = \sin x$



Area of OAB

$$\begin{aligned} &= \int_0^\pi y dx = \int_0^\pi \sin x dx = [-\cos x]_0^\pi \\ &= -[\cos \pi - \cos 0] = -(-1 - 1) = 2 \text{ sq. units} \end{aligned}$$

Q. For function $y = 2 \sin \theta$

Find average of y from $\theta = 0$ to $\theta = \pi$.

Sol. $y_{\text{avg}} = \frac{2 \int_0^\pi \sin \theta d\theta}{\pi - 0} = \frac{2}{\pi} [-\cos \theta]_0^\pi = \frac{4}{\pi}$

Q. Particle's position as a function of time is given by $x = -t^2 + 4t + 4$, find the maximum value of position co-ordinate of particle.

Sol. $x = -t^2 + 4t + 4$

$$\frac{dx}{dt} = -2t + 4 = 0$$

$$t = 2$$

$$x = -(2)^2 + 4(2) + 4$$

$$x = 8$$

Q. Find minimum value of the function: $y = 25x^2 + 5 - 10x$

Sol. $\frac{dy}{dx} = 50x - 10 \Rightarrow \frac{dy}{dx} = 0 \Rightarrow x = \frac{1}{5}$

$$\frac{d^2y}{dx^2} > 0$$

$$\therefore y_{\text{min}} = 4$$

Q. Determine the position where potential energy will be minimum if $U(x) = 100 - 50x + 1000x^2$.

Sol. $U(x) = 100 - 50x + 1000x^2$

$$\frac{dU}{dx} = -50 + 2000x = 0 \Rightarrow x = \frac{1}{40}$$

$$x = 2.5 \times 10^{-2}$$

Q. Find out minimum/maximum value of $y = 4x^2 - 2x + 3$ also find out those points where value is minimum/maximum.

Sol. $y = 4x^2 - 2x + 3$

$$\frac{dy}{dx} = 8x - 2 = 0$$

$$x = \frac{1}{4} \quad y = 4\left(\frac{1}{4}\right)^2 - 2\left(\frac{1}{4}\right) + 3 = \frac{1}{4} - \frac{2}{4} + 3$$

$$y = \frac{11}{4} \Rightarrow \frac{d^2y}{dx^2} = 8 > 0 \rightarrow \text{minimum}$$

Q. $x + y = 10$ then find the maximum value of $f = xy$?

Sol. $f = x(10 - x) = 10x - x^2$

$$\frac{df}{dx} = -2x + 10 = 0$$

$$x = 5$$

$$\frac{d^2f}{dx^2} = -2$$

max value at $x = 5, y = 5$

$$f = xy = 25$$

Q. If $\ell + r = 12$ here ℓ is length of cylinder and r is radius of cylinder then find maximum value of volume of cylinder

Sol. Volume of cylinder $= \pi r^2 \ell$

$$V = \pi r^2(12 - r)$$

$$V = \pi(12r^2 - r^3)$$

$$\frac{dV}{dr} = \pi[12(2r) - 3r^2] = 0$$

$$\frac{dV}{dr} = \pi[24r - 3r^2] = 0$$

$$24r = 3r^2$$

$$r = 8$$

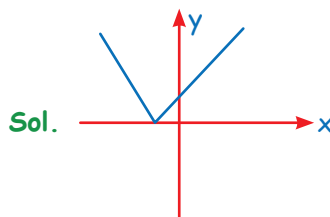
$$\frac{d^2V}{dr^2} = \pi(24 - 6r)$$

$$= \pi(24 - 48) = -24\pi \text{ max.}$$

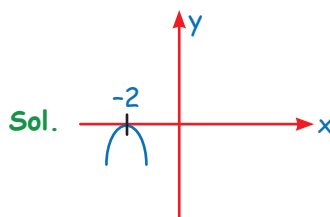
$$\therefore \text{max value of } V = \pi \times 64 \times 4 = 256\pi$$

COORDINATE GEOMETRY & GRAPHS

Q. Correct graph of $y = |3x + 4|$ is:

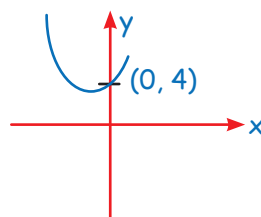


Q. Correct graph of $y = -(x + 2)^2$ is:



Q. Graph of $y = 2(x + 1)^2 + 2$ is:

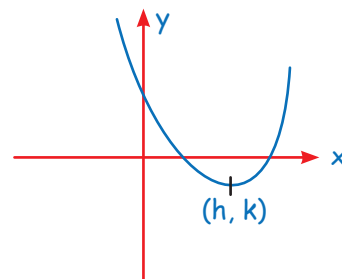
Sol. Displaced point $= (-1, 2), a > 0$



Q. Graph of $y = 3x^2 - 4x + 1$ is:

Sol. $h = -\frac{b}{2a} = \frac{4}{2 \times 3} = \frac{2}{3}$

$$k = -\left(\frac{b^2 - 4ac}{4a}\right) = -\left(\frac{16 - 4 \times 1 \times 3}{4 \times 3}\right) = -\frac{1}{3}$$



Q. Find the slope of the line $-3x + 4y = 11$.

Sol. $-3x + 4y = 11$

$$4y = 11 + 3x$$

$$y = \frac{11}{4} + \frac{3x}{4} \Rightarrow c = \frac{11}{4}, m = \frac{3}{4}$$

Q. The equation of straight line having slope $\sqrt{3}$ and y intercept of -2 will be:

Sol. $y = mx + c$

$$y = \sqrt{3}x - 2$$

Q. The equation of straight line is $2y = 3x - 6$ its x and y intercepts are

Sol. $y = \frac{3}{2}x - 3$

To get y - intercept, x term = 0

To get x - intercept, y term = 0

$$x = 2$$

$$y = -3$$

Q. The distance between points (2, 3, -7) and (-2, 0, 5) is

Sol. Distance b/w two points (x_1, y_1, z_1) & (x_2, y_2, z_2) is

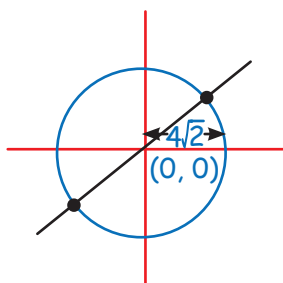
$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$= \sqrt{(-2 - 2)^2 + (0 - 3)^2 + (5 + 7)^2}$$

$$= \sqrt{16 + 9 + 144} = \sqrt{169} = 13$$

Q. The points where the line $y = x$ intersect the curve $x^2 + y^2 = 32$ is

Sol.



$x = y$ eqⁿ of straight line

$x^2 + y^2 = (4\sqrt{2})^2$ eqⁿ of circle of centre (0, 0),

radius $4\sqrt{2}$

St. line intersect circle at two points $(+a, +a)$ and $(-a, -a)$

$$x^2 + x^2 = 32$$

$$2x^2 = 32$$

$$x = \pm 4$$

$\therefore$ st line intersect circle at point $(+4, +4)$ & $(-4, -4)$

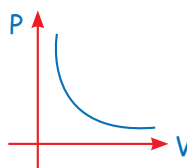
Q. Graph of $PV = \text{constant}$ is:

Sol. $PV = nRT$

at constant temp

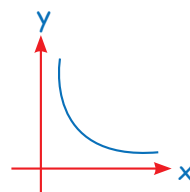
$$PV = \text{const.}$$

$$P \propto \frac{1}{V}$$



Q. Graph of $x^2y = 2$ is:

Sol. $x^2y = 2$ graph is;



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